

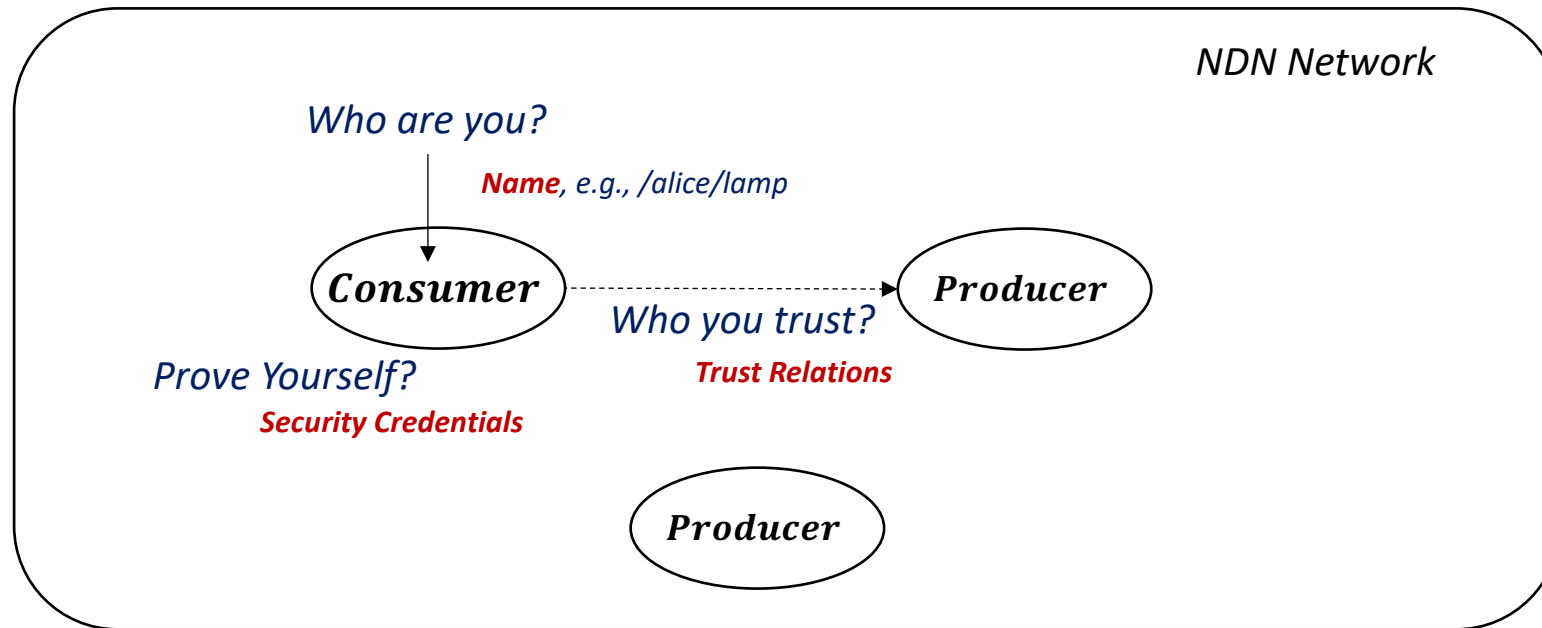
Bootstrapping Remote NDN Entity Leveraging Existing Authentications

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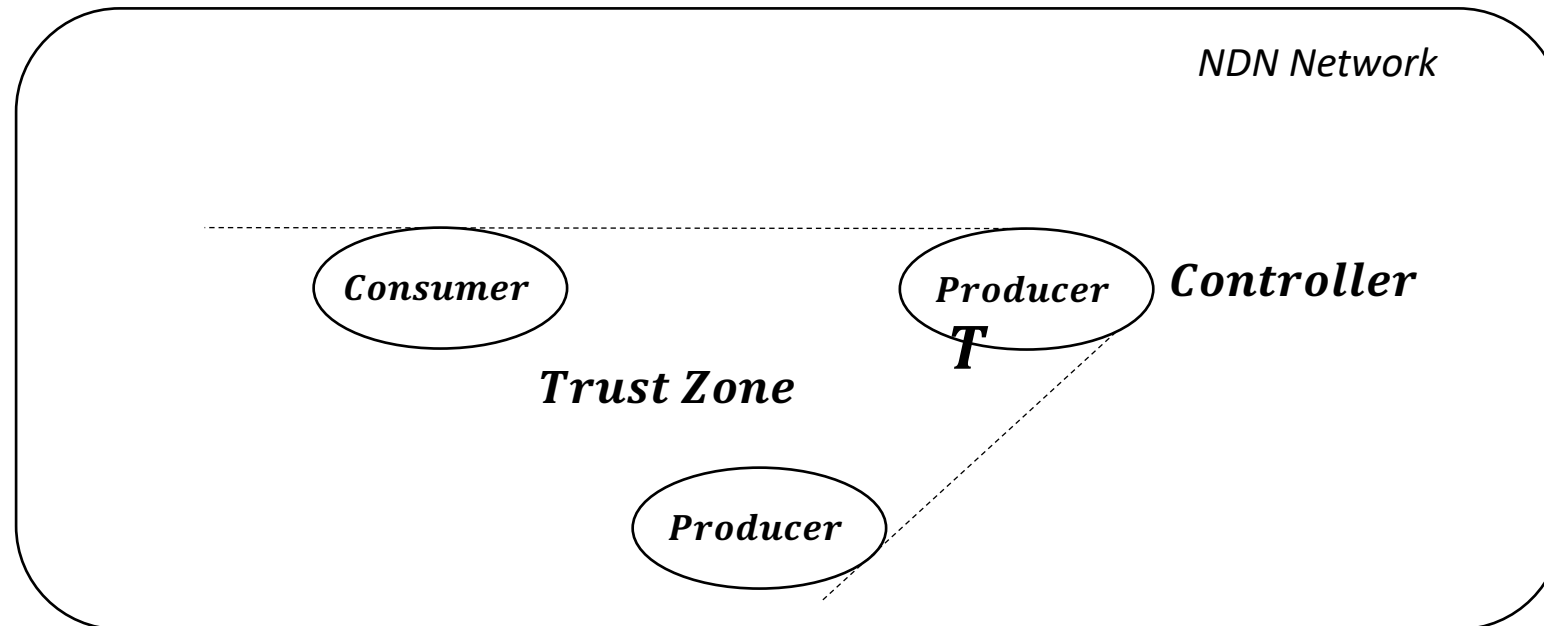
NDN Entity Needs Configuration

- NDN claims data are secured, but...
 - Consumers/Producers need configurations to enable secure communications
 - *Name, security credentials, and trust relations*



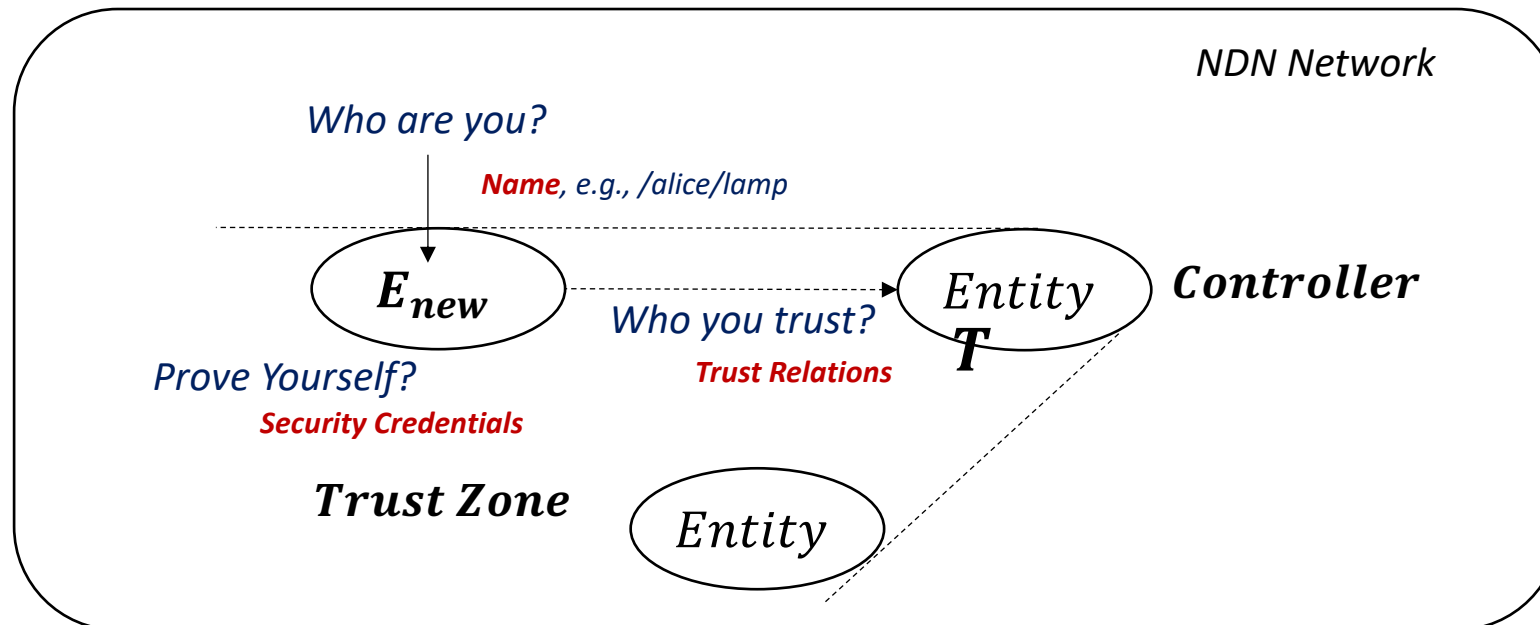
Terminologies

- NDN entity can create a trust anchor ***T***
- All NDN entities under the same trust anchor make ***Trust Zone***
- Owner of the trust anchor ***T*** is the ***Controller*** of this trust zone



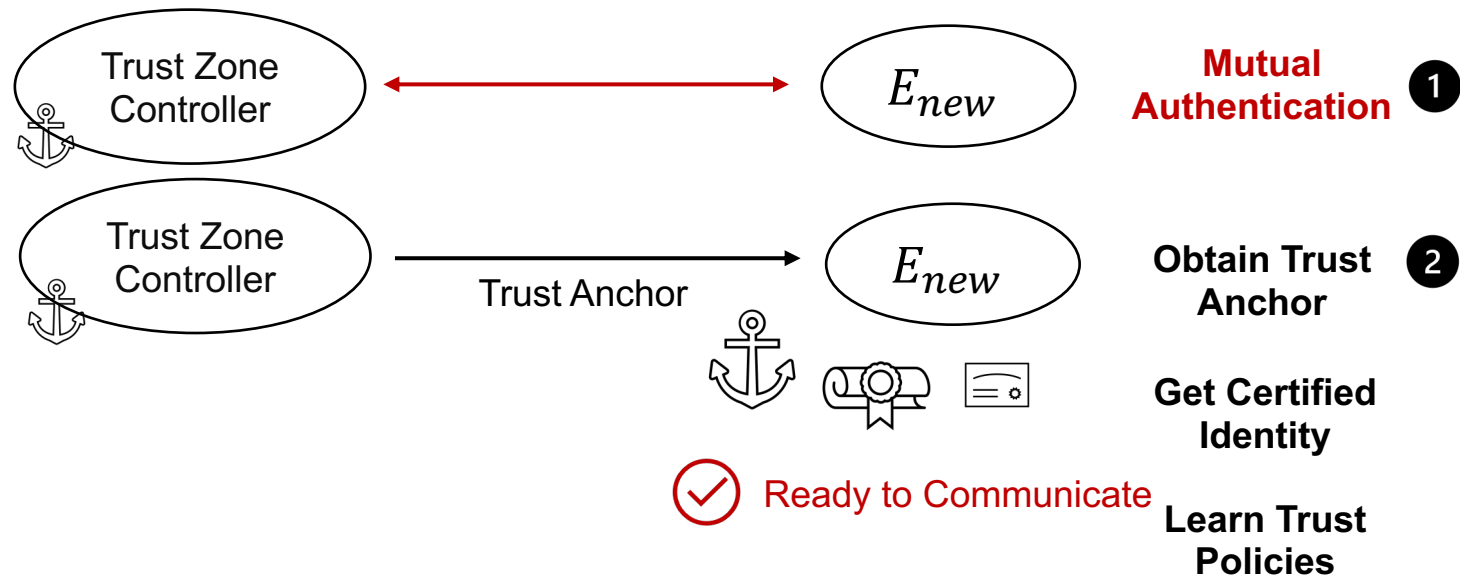
Security Bootstrapping

- Bootstrapping/Configuring a new entity E_{new} is
 - Configuring E_{new} into a “trust zone”
 - E_{new} must have a **name**, obtained on its own, or otherwise assigned by the “Trust Zone Controller”
 - E_{new} must have its **trust anchor**, **certificate** and **trust policies** installed
 - We refer this configuration process as security bootstrapping



Security Bootstrapping (contd.)

- Logical steps of security bootstrapping



Existing Works

- NDNCERT
 - Issuing certificate only, leaving authentication methods as a need to be filled
- Data-Centric Toolkit (DCT)
 - Command line tools to generate “security bundle”
 - Out-of-band installing the bundle into E_{new}
 - Mutual authentication through “local, physically connected” environment
- Need to solve new problems: remote bootstrapping
 - Hydra could be a good case study

Bootstrapping Remote E_{new}

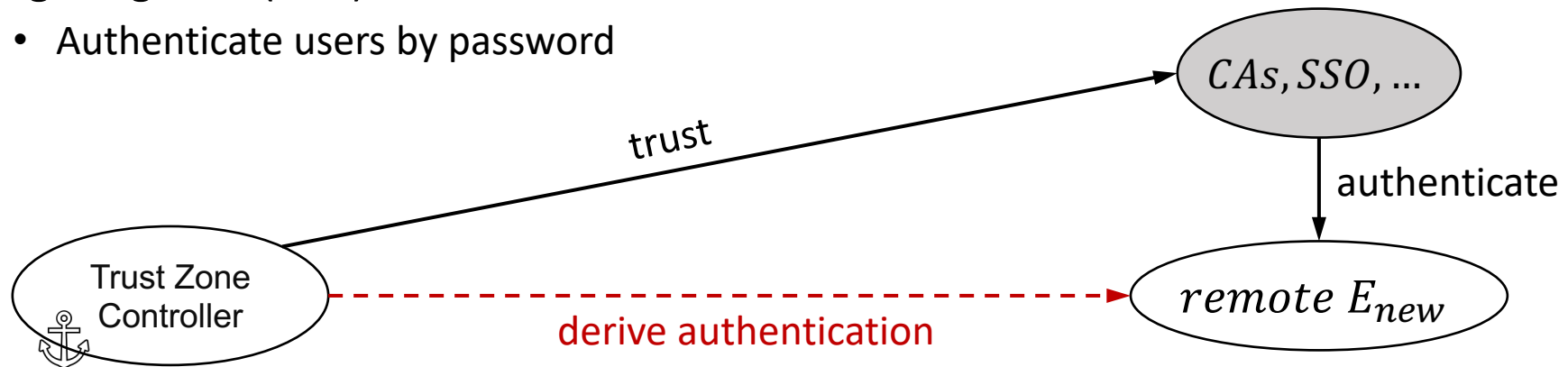
- Mutual authentication: *needs factors to verify, where they are from?*
 - Answers depend on use case scenarios
- We developed specific security bootstrapping processes for the remote case
- E_{new} is remote (i.e., an application on remote server)
 - Trust Zone Controller can only reach remote E_{new} **over unsecured TCP/IP connectivity**



Achieve **mutual authentication**, then **configure**
trust anchor, name, certificate, trust policies

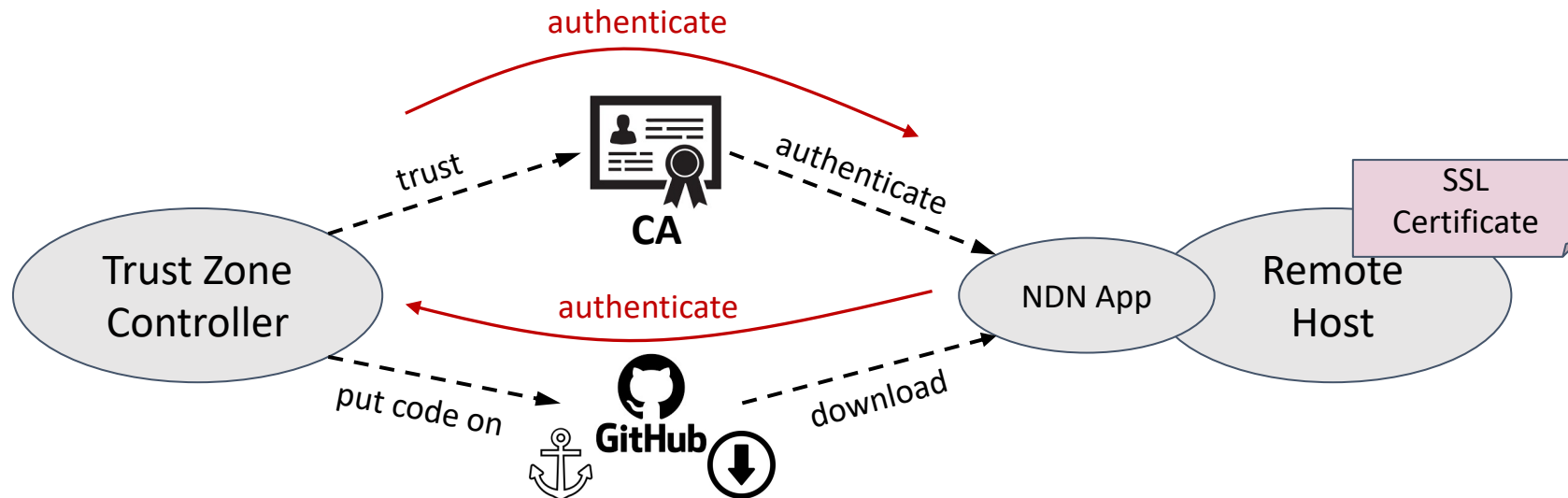
Leveraging Existing Authentications

- To achieve mutual authentication between Trust Zone Controller and E_{new}
 - We look into leveraging **existing authentications solutions**
- Multiple such solutions exist in today's Internet
 - Certificate Authority (CAs), DNSSEC
 - Authenticate servers by their DNS names
 - Single Sign-On (SSO)
 - Authenticate users by password



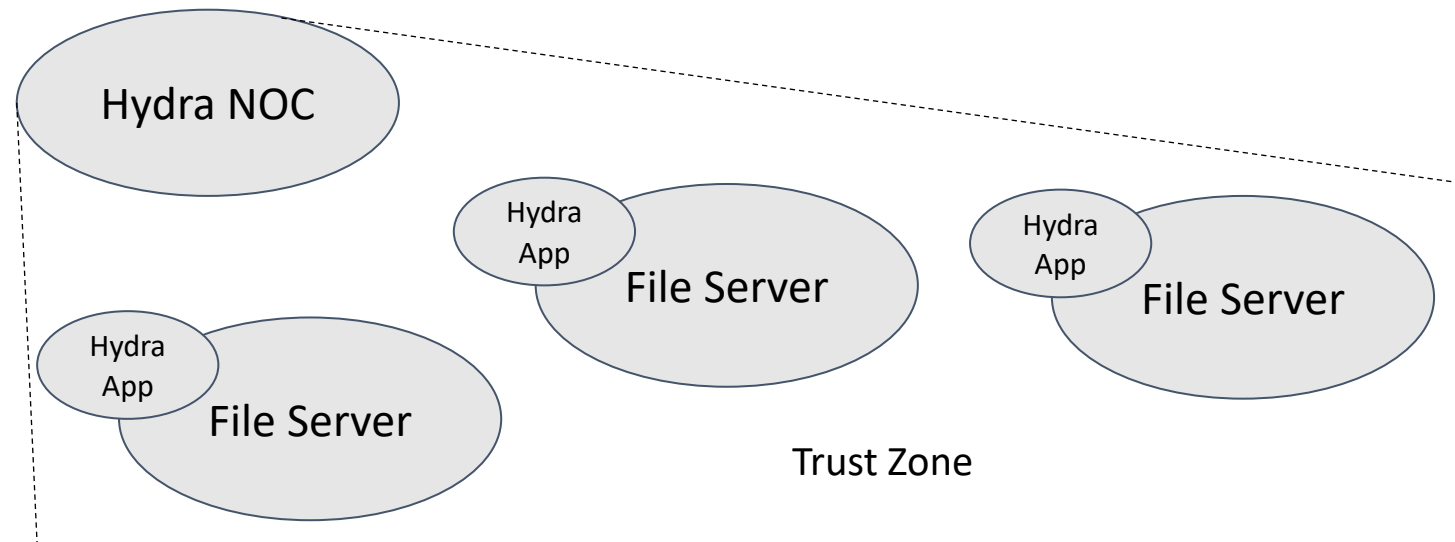
Achieving Mutual Authentication

- Assuming E_{new} is an NDN app running on remote host H
- **Trust Zone Controller authenticates E_{new}**
 - If the DNS name of H is authenticated by CA (i.e., SSL/TLS Certificate)
- **E_{new} authenticates Trust Zone Controller**
 - Built-in during software distribution: trust anchor embedded



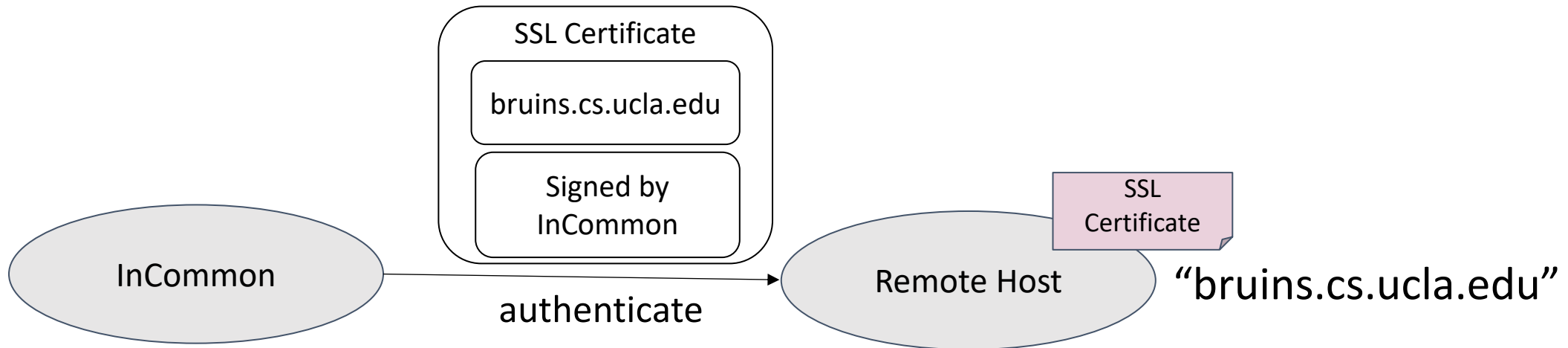
Bootstrapping remote Hydra app

- Organizations contribute file server and share data
 - Users install Hydra app on contributed file servers
 - Hydra Networking Operating Center (NOC) serves as Trust Zone Controller for “/hydra”
- The Hydra app on remote hosts need authentication



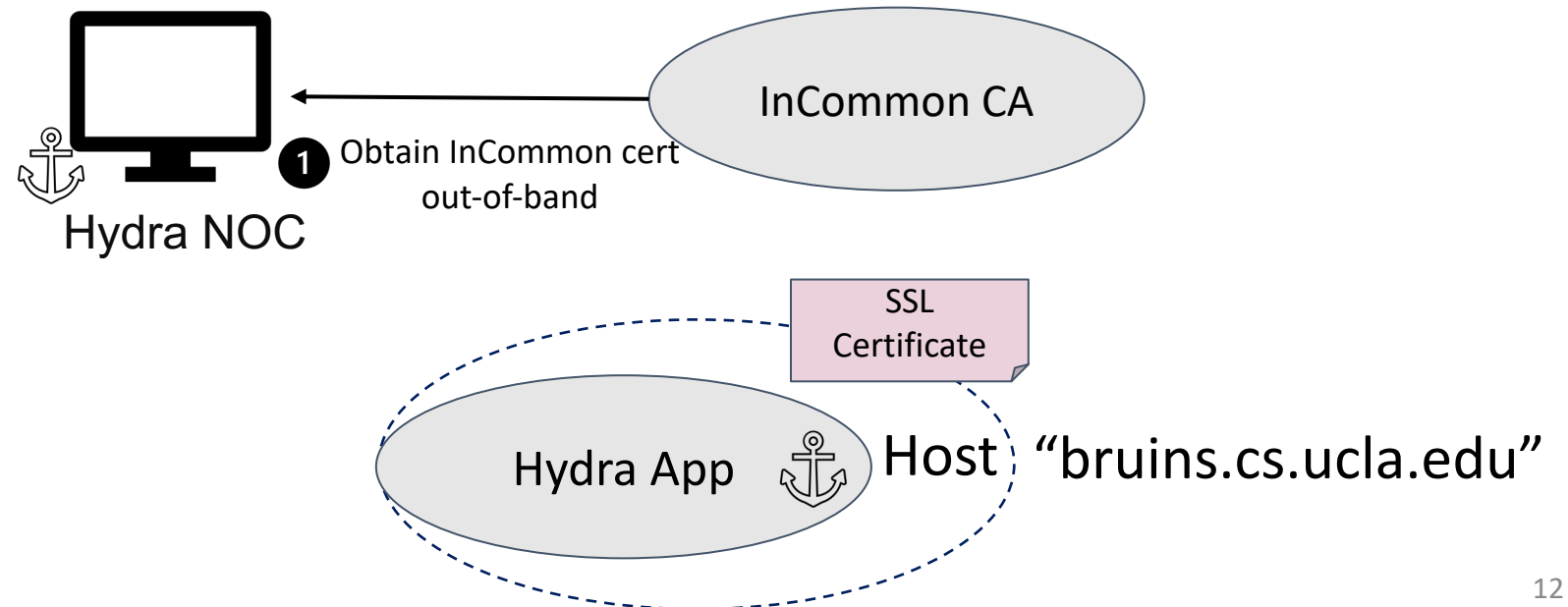
Authenticating Remote Host via SSL Certificate

- The remote host already has an assigned, CA authenticated host identifier
 - For example, DNS name *“bruins.cs.ucla.edu”*
 - UCLA’s CA InCommon authenticates it by SSL certificate
- The SSL certificate is the remote host “existing” authentication



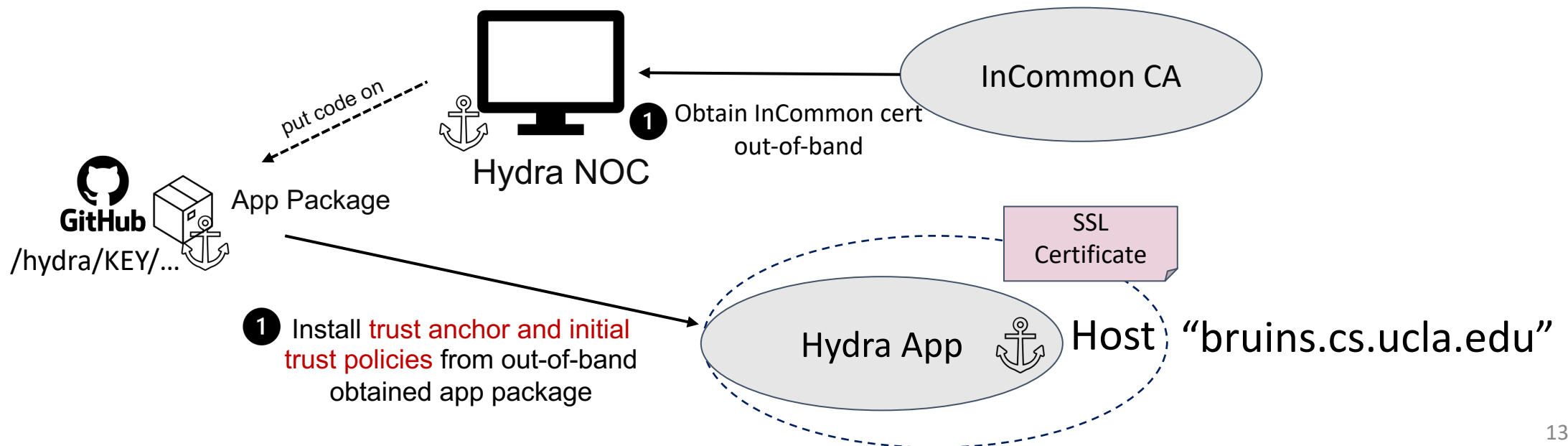
Authenticating E_{new}

- Hydra NOC trusts InCommon CA out-of-band
 - *e.g.*, download InCommon CA certificate from trusted URL
- Hydra NOC can authenticate all InCommon authenticated host identifiers
 - Thereby can **authenticate Hydra app instances** installed on them



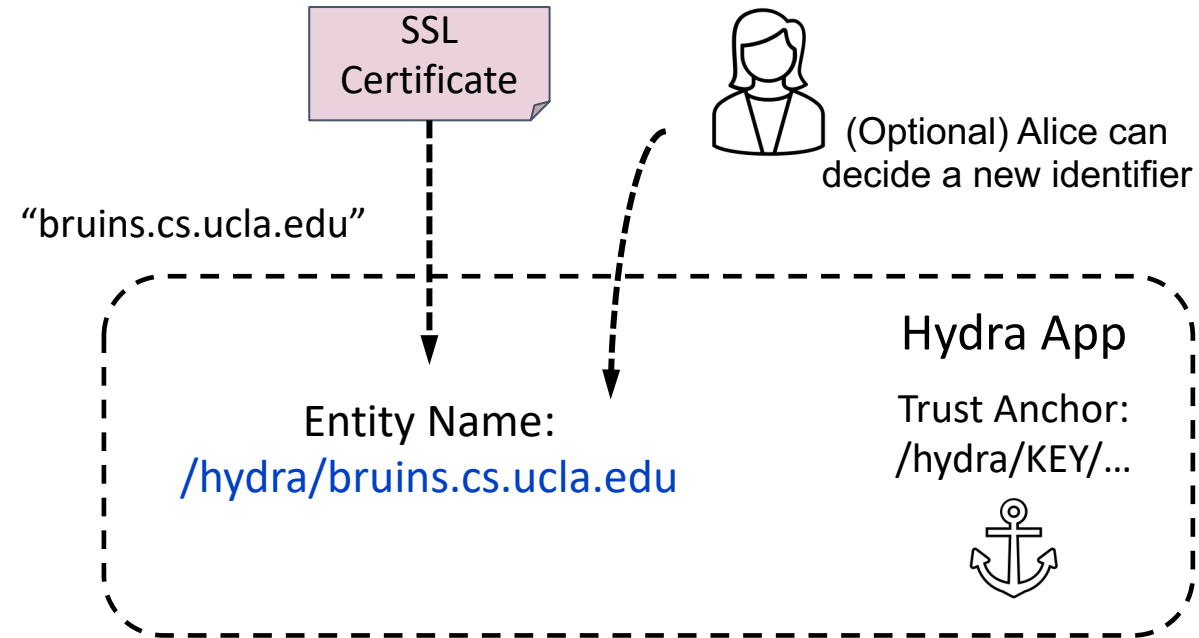
Authenticating Trust Zone Controller

- **Hydra app authenticates Hydra NOC at the application installation time**
 - Hydra trust anchor and initial trust policies are embedded in the app package that implements the Hydra app
 - User (e.g., Alice) authenticates app package out-of-band (e.g., Github URL)
 - *Critical aspect: URL must be obtained from trusted source*



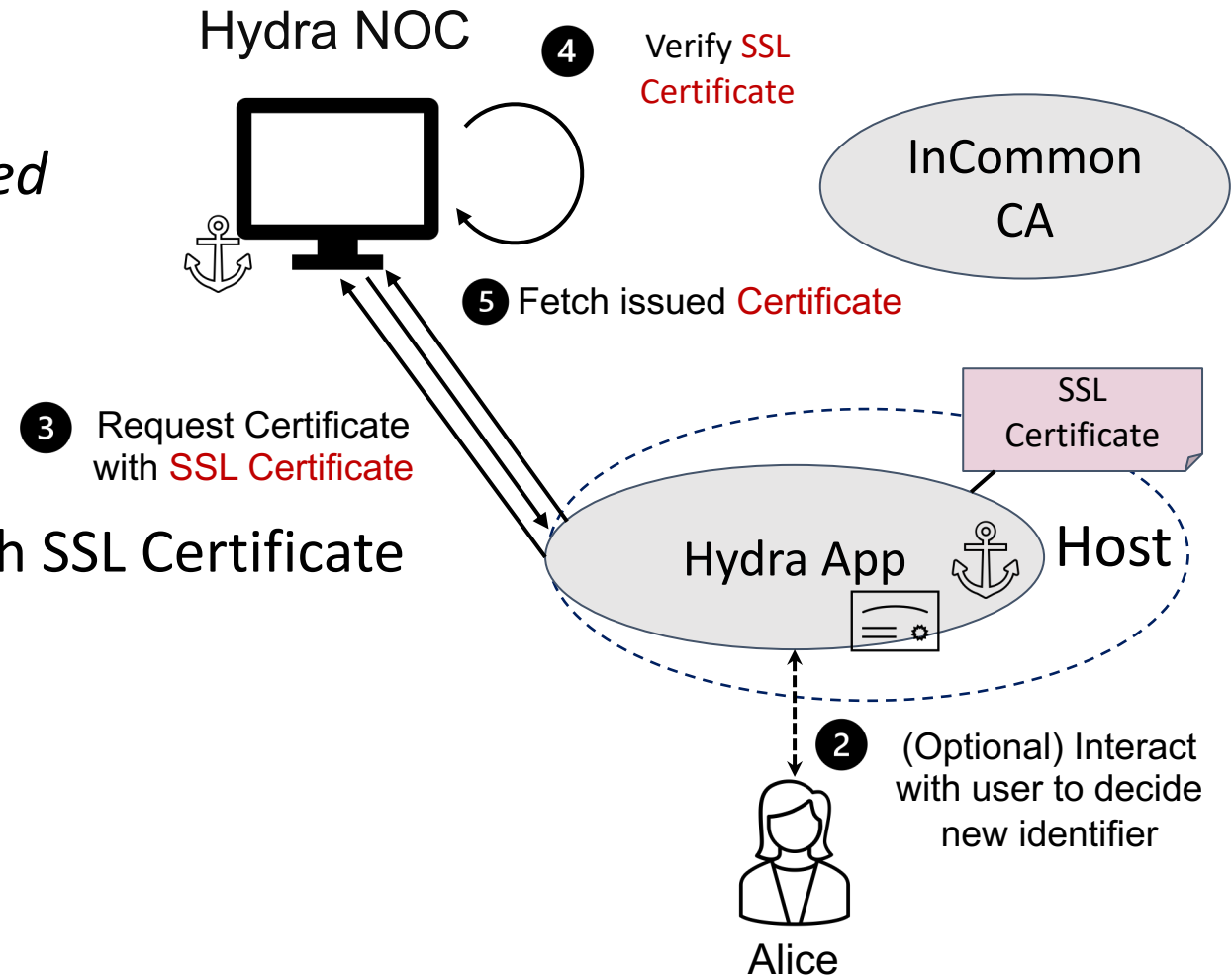
Naming Remote E_{new}

- E_{new} name has an application prefix and unique suffix
 - Application prefix comes from trust anchor
 - Unique suffix needs assignment
- Give Hydra app a “new” name
 - A name in hydra’s context
 - e.g., “/hydra/bruins.cs.ucla.edu”
 - Optionally, Alice can decide a new identifier



Obtaining Certificate

- *Mutual authentication is achieved*
- *Trust anchor, initial trust policies installed*
- *Name is self-obtained*
- **Certificate still needed**
- Hydra app uses NDNCERT protocol to apply certificate from Hydra NOC with SSL Certificate



Demo

Next Step

- Further implementation of Hydra bootstrapping
- New scenarios in mGuard, NEAR...